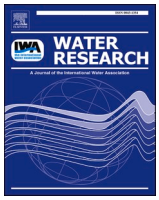
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Predicting and understanding the performance of polyamide nanofiltration membrane for Li/Mg selective separation based on machine learning

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ABSTRACT

Nanofiltration technology holds significant promise for the selective separation of monovalent and multivalent ions, such as lithium (Li) and magnesium (Mg), during Li extraction from salt lakes. Nevertheless, optimizing polyamide nanofiltration membranes for selective ion separation remains inherently challenging due to the complex interactions between ions and membrane structural and experimental parameters, making the ion transport mechanism ambiguous. This work employed a machine learning (ML) approach to identify and comprehensively understand the features that influence the membrane permeability and selectivity using a comprehensive dataset, which encompassed fabrication parameters, experimental conditions, membrane properties, and single salt rejection performance. Initially, ML algorithms accurately predicted intrinsic membrane properties using only fabrication parameters but struggled to predict permeation and selectivity when combining fabrication parameters or membrane properties with experimental conditions. To address this limitation, salt rejection performance was incorporated, and various combinations of input variables were systematically compared to identify optimal input configurations for robust ML algorithms capable of accurately predicting membrane permeability and selectivity. Using the Shapley additive explanation (SHAP) method, we found that membrane permeability was mainly determined by fabrication parameters such as substrate type and heat curing temperature. While these parameters also influenced molecular weight cut-off (MWCO) and zeta potential, they did not fully reflect the physicochemical factors governing ion separation. In contrast, $MgCl_2$ rejection served as a more integrative and informative descriptor, capturing both pore structure and surface electrostatic effects in predicting Li/Mg selectivity. These findings underscore the necessity of a multifaceted approach in modeling membrane performance, integrating both intrinsic properties and external factors to achieve optimal ion selectivity.

1. Introduction

Global demand for lithium (Li)—a critical raw material in Li-ion batteries powering the clean energy transition—is surging, with production projected to increase by 18–20-fold by 2050 (Vera et al., 2023; Yong et al., 2024a). Currently, the primary global sources of Li are sourced from salt-lake brine (62.6%) and hard-rock ores (37.4 %) (Ding et al., 2023). Li extraction, a critical industrial challenge (Li et al., 2019; Xu et al., 2021), from salt-lake brine has proven to be more economical and sustainable compared to from hard rock (Paranthaman et al., 2017; Yong et al., 2024a). However, high Mg-to-Li ratio (MLR) of brine poses great challenge to traditional methods like evaporation and chemical precipitation (Wang et al., 2023). Integration membrane processes with traditional methods enable Li extraction from brine with a high MLR. Among them, the nanofiltration (NF) process—being pivotal and technically challenging—govern the separation Li and Mg from brine with low-energy input (Li et al., 2019). Therefore, in recent years, numerous efforts have been made to develop high-performance NF membranes for Li/Mg selective separation.

Polyamide-based thin-film composite (TFC) NF membranes are the concerned technology in Li/Mg selective separation due to their successfully commercialized in water purification (Peng et al., 2024; Shen et al., 2022). The Li/Mg selective separation of a polyamide-based TFC NF membrane is primarily governed by polyamide active layer on the

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basis of Donnan exclusion, dielectric exclusion and steric hindrance (Hu et al., 2018; Lu et al., 2021; Zhai et al., 2024). These separation mechanisms are closely related to surface properties of polyamine layer, including thickness, charge, hydrophilicity, and pore structure, etc., which have been carefully regulated to enhance Li/Mg selectivity in polyamide-based TFC NF membranes. For instance, increasing surface positive charge and creating negatively charged membrane surface can efficiently separation Li from solution with high MLR (Guo et al., 2024a; Li et al., 2017, 2015; Peng et al., 2023, 2024; Xu et al., 2023; Zhang et al., 2017). Other studies reported the importance of pore size distribution of polyamide-based TFC NF membranes in the Li/Mg selective separation (Liang et al., 2020; Peng et al., 2024; Yuan et al., 2024). Additionally, external factors, encompass experimental conditions (such as MLR of feed solution, concentration of solution, and filtration pressure, etc.), significantly affected the Li/Mg selectivity of polyamide-based TFC NF membranes (Guo et al., 2024b; Li et al., 2023). Collectively, a diverse array of variables governs the Li/Mg separation efficiency of polyamide-based TFC NF membranes. Therefore, a comprehensive understanding of the underlying structure-performance relationship and the mechanisms driving Li/Mg selectivity is essential for the rational design of purpose-specific polyamide-based TFC NF membranes.

Machine learning (ML), as a typical data-driven method, is powerful for mining complex and multidimensional data to reveal the relationship among various variables. In this regard, ML have shown substantial potential in membrane separation field, such as predict membrane performance, identify key variable for improving membrane design and elucidate separation mechanism, etc. (Gao et al., 2023; Hu et al., 2025; Jeong et al., 2021, 2023; Lu et al., 2023; Ritt et al., 2022; Wang et al., 2024; Yang et al., 2023). For example, recent studies have explored diverse applications including unraveling solvent effects in organic solvent nanofiltration (Ignacz et al., 2023), predicting flux in membrane bioreactors (Liu and Liao, 2025), modeling energy and carbon costs in separation systems (Ignacz et al., 2025), and forecasting solute rejection via molecular fingerprints (Lee et al., 2024). To date, ML algorithms have rarely been applied to predict Li/Mg selective separation of polyamide nanofiltration membranes (Lu et al., 2023; Yong et al., 2024b; Zheng et al., 2024). Additionally, existing studies primarily focus on the impact of membrane properties and experimental conditions on permeation and selectivity, often neglecting the role of fabrication parameters. This oversight is pivotal, as fabrication conditions are closely linked to membrane properties and directly affect separation performance (Barnett et al., 2020; Yang et al., 2022). As a result, previous studies have inadequately addressed the interdependencies among multiple variables, failing to elucidate the complex interplay between membrane fabrication, membrane properties, and membrane performance. Therefore, there is a pressing need to develop comprehensive ML frameworks.

In this work, we developed ML models to predict the performance of polyamide TFC NF membranes and investigate the factors influencing their Li/Mg separation efficiency. A comprehensive dataset was extracted from the literature, including fabrication parameters, experimental conditions, membrane properties, and single salt rejection performance. By integrating intrinsic membrane properties with experimental conditions, the developed ML models achieved accurate predictions of membrane permeability but exhibited limited selectivity predictions. Incorporating salt rejection capability alongside these factors significantly enhanced selectivity predictions ($R^2 = 0.89$, Spearman coefficient = 0.91). Shapley additive explanations (SHAP) analysis revealed that fabrication parameters—including substrate membrane type (specifically polysulfone) and amine concentration—and membrane thickness significantly influence permeation properties, while selectivity was most affected by $MgCl_2$ rejection and amine monomer type. The superior performance of ML models in predicting selectivity using comprehensive input features highlights the complexity of selective ion transport mechanisms and the importance of detailed membrane characterization.

2. Materials and methods

2.1. Data collection

Data for Li/Mg separation using polyamide TFC NF membranes were collected from 59 peer-reviewed publications identified using the keywords “polyamide nanofiltration membranes” and “Li/Mg separation” in Google Scholar (https://scholar.google.com/). A total of 150 data points was extracted from these publications and used for subsequent analysis and modeling. During the extraction, we considered input variables including fabrication parameters (36 variables), membrane properties (7 variables), experimental conditions (7 variables), and single salt rejection performance (2 variables) (Table S1). When these variables were not explicitly mentioned in the text, the "Engauge Digitizer" tool was used to extract their values from accompanying graphs. The complete dataset is available in the Supporting Information.

2.2. Data preprocessing

Subsequently, the missing values in the collected data, which were caused by the difference of fabrication conditions, experimental conditions, and measured properties in the various publications, were preprocessed for the convenience of modeling. Specifically, the features contained >40 % missing values were removed from the dataset. For the remaining features (Table S2), we filled in the missing values according to certain rules. For continuous-value features, we filled it using the average value of that feature. For categorical features, we treated missing values as an additional class. Additionally, the class variables were converted to numerical variables based on one-hot encoding. It is noting that the preprocessing was solely for ML model training purposes and did not imply that we disregarded the influence of other features. Possible effect of preprocessing on the model performance and interpretation was further discussed in the Supplementary Materials.

2.3. Model construction and evaluation

For each target membrane property or Li/Mg separation performance, a model was constructed and evaluated following a standardized workflow, which included input feature selection, dataset partitioning, algorithm selection, hyperparameter tuning, model training, and performance evaluation. For membrane property prediction, only fabrication parameters (as listed in Table S2) were used as input features. In contrast, the model for Li/Mg separation performance employed an optimized combination of four types of input features: fabrication parameters, membrane properties, experimental conditions, and single salt rejection data. The dataset was randomly divided into training and testing set at the ratio of 8:2, with the training set used for model development and the test set reserved strictly for final performance evaluation to prevent data leakage.

Eleven commonly used regression algorithms were initially tested using default hyperparameters, including linear regression (LR), ridge regression (Ridge), least absolute shrinkage and selection operator (LASSO), support vector regression (SVR), multilayer perceptron (MLP), k-nearest neighbors (KNN), decision tree (DT), and four tree-based ensemble models: AdaBoost (Ada), Extremely Randomized Trees (ET), Bagging (Bag), and Random Forest (RF). All these models were implemented using the scikit-learn library. Algorithm selection was based on five-fold cross-validation performance on the training set. The best-performing model was further optimized via grid search over key hyperparameters, then retrained on the entire training set to construct the final predictor. Possible effect of hyperparameters on the model performance and interpretation was further discussed in the Supplementary Materials.

Model performance was evaluated on the test set using four standard

regression metrics, including mean absolute error (MAE, Eq. (1)), root-mean-square error (RMSE, Eq. (2)), the coefficient of determination (R^2, Eq. (3)), and Spearman's coefficients (ρ, Eq. (4)). MAE and RMSE quantify the discrepancies between predicted and observed values, while Spearman's coefficients (ρ) and R^2 assesses the goodness of fit. Lower values of MAE and RMSE combined with a higher Spearman's coefficients and R^2 signify better predictive performance. To ensure the robustness and reliability of the performance estimates, such an entire process (random train-test split → cross-validation → model training → test evaluation) was repeated five times with different random seeds. The final reported performance metrics represent the average across these five independent runs.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \in [0, +\infty) \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \in [0, +\infty) \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2} \in [0, 1] \quad (3)$$

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \quad (4)$$

where n represents the number of samples, y refers the real values, and $\hat{y}$ refers the predicted values, d denotes the ranking difference.

2.4. Model interpretation

The model interpretation based on Shapley additive explanations (SHAP), a widely-used post-hoc interpretation algorithm, was carried out to enhance credibility and reveal the knowledge embedded in the model. The algorithm can calculate the influence of each input feature on the output for every sample. Based on the results, the feature importance and the correlation between input features and target properties were obtained and analyzed, to provide guidance for future membrane design.

3. Results and discussion

3.1. Statistical analysis of the dataset

To gain a deeper understanding of the data and uncover underlying

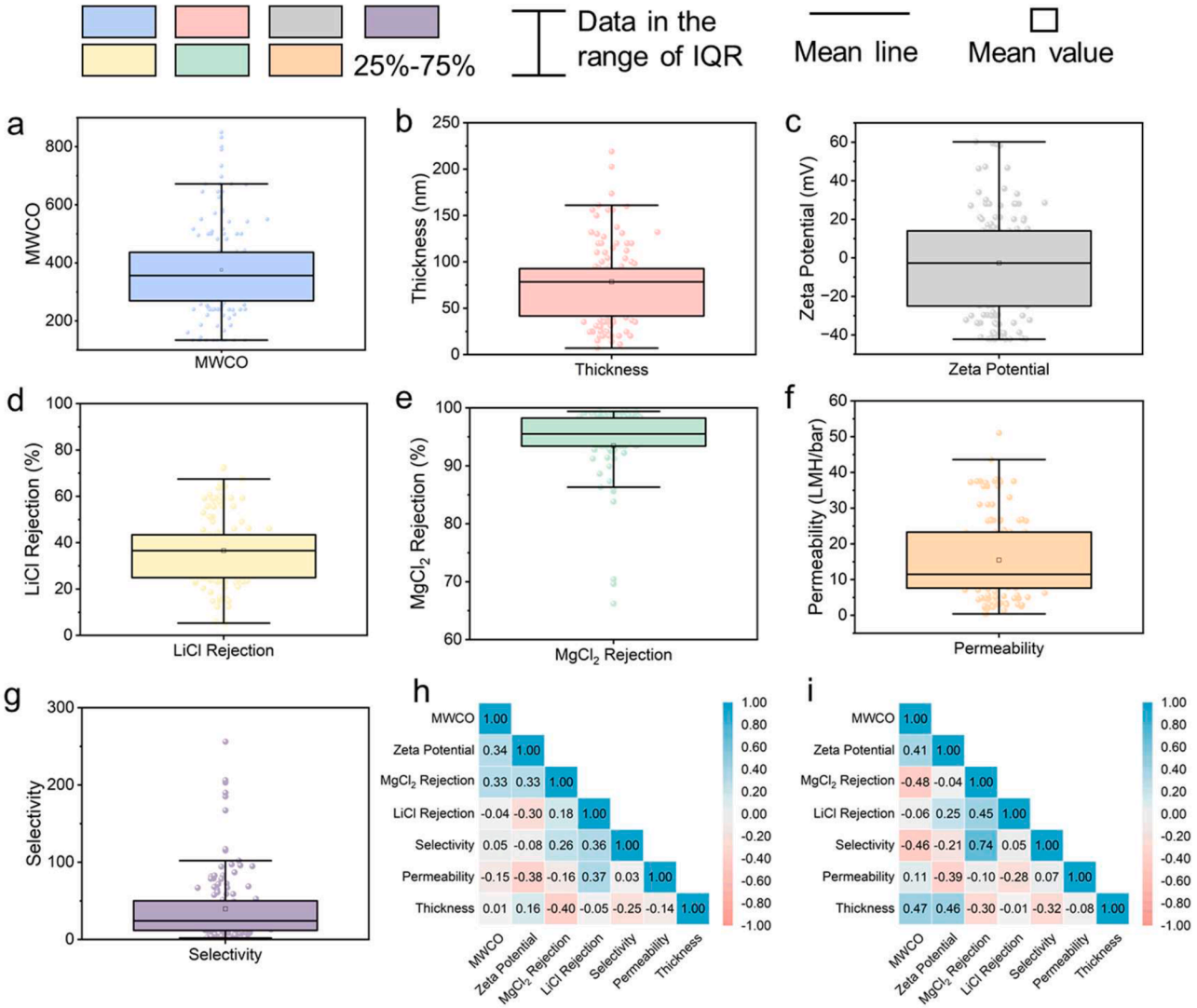


Fig. 1. (a-g) Statistical analysis of the target features of membrane properties and performance, including (a) MWCO, (b) thickness, (c) zeta potential, (d) LiCl rejection, (e) $MgCl_2$ rejection, (f) permeability, and (g) selectivity. (h-i) The (h) Pearson and (i) Spearman correlation matrix of the target features.

patterns for more informed modeling decisions, a statistical analysis was conducted on the dataset. Fig. 1a-g display the data distribution of the target variables (i.e., molecular weight cut-off (MWCO), thickness, zeta potential, $MgCl_2$ rejection, LiCl rejection, membrane permeability, and Li/Mg selectivity). The major data regarding membrane property and performance features are highlighted in the box plot, where the MWCO dataset between the 25 % and 75 % was located in a range of 270 to 437, correlating with a zeta potential between −25 mV and +14 mV, and membrane thicknesses ranging from 41.7 nm to 92.7 nm. Regarding membrane performance, the rejection of $MgCl_2$ ranges from 93.4 % to 98.3 %, while LiCl rejection spans from 24.9 % to 43.4 %. The permeability dataset predominantly falls between 7.61 LMH/bar and 23.3 LMH/bar, and the Li/Mg selectivity varies from 11.7 to 50.2. Notably, most variables exhibit a long-tail distribution, with sparse data near the maximum and minimum values. In such cases, the data splitting method can significantly impact model performance. For example, if data from the tail regions only appear in the test set, the model may struggle to make accurate predictions. Therefore, for each target property, five random data splits (by altering the random seed) were performed, and the model was evaluated in all five split sceneries to accurately reflect its performance.

Pearson correlation coefficients, which assess the strength of linear correlations between pairs of variables, were calculated to evaluate the relationships among the selected features. As shown in Fig. 1 h, the Pearson correlation coefficients ranged from −0.40 to 0.37, indicating negligible correlations between the features and thereby ensuring robustness of subsequent model interpretations and applications. Additionally, Spearman correlation, which measures monotonic relationships between variables, aids in identifying relevant features and enhancing model prediction accuracy. As depicted in Fig. 1i, selectivity displayed a strong monotonic correlation with $MgCl_2$ rejection (Pearson coefficient = 0.77), highlighting the significant relationship between $MgCl_2$ rejection and selectivity. Incorporating these correlations into machine learning models can improve the accuracy of predictions for the target variables.

3.2. Performance of developed ML model for membrane properties prediction

Membrane performance, encompassing both permeability and selectivity, has been proved to be affected by intrinsic membrane properties such as molecular weight cut-off (MWCO), thickness, and zeta potential. These variables are widely used to predict membrane separation performance; however, the relationship between these properties and membrane synthesis conditions has yet to be established in the context of polyamide membranes for Li/Mg separation, hindering direct guidance for the synthesis of high-performance membranes. Therefore, we first developed a model to accurately predict the intrinsic properties of membranes based on synthesis conditions.

The optimal ML algorithm was identified from eleven commonly-

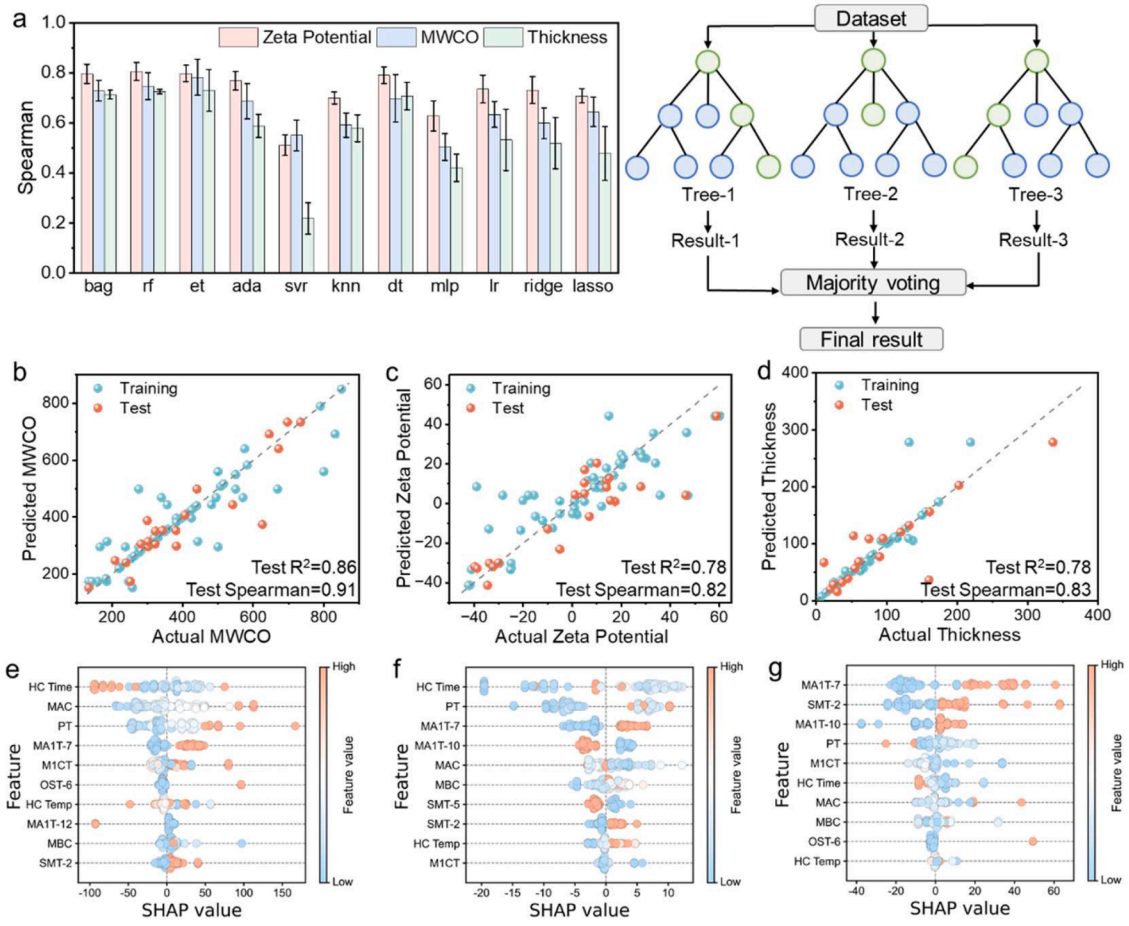


Fig. 2. (a) Evaluation of the predictive performance of eleven ML algorithms for MWCO, zeta potential, and thickness. The tree-based model is schematically illustrated on the right side of Fig. 1a. (b-d) Prediction performance of the developed models for (b) MWCO, (c) zeta potential, and (d) thickness based on fabrication parameters. All five random seed plots for the prediction performance are provided in Figs. S1–3. (e-g) SHAP summary plot of the ET model for (e) MWCO, the RF model for (f) zeta potential, and the ET model for (g) thickness. HC Time, MAC, PT, HC Temp, MBC, and M1CT represent heat curing time, monomer A concentration, polymerization time, heat curing temperature, monomer B concentration, and monomer A contact time, respectively. MA1T-number, OST-number, SMT-number denote the encoders for the categorical feature of the type of monomer A1, the type of organic solvent, and the type of substrate membrane, respectively.

used ones for each target variable (i.e., MWCO, thickness, and zeta potential). Table S3 presents the Spearman correlation coefficients, as well as the MAE and RMSE values for each model, derived from the source data for the predictions of these properties. Specifically, algorithms such as SVR, MLP, KNN, and RIDGE exhibited Spearman correlation coefficients with relatively low means and large variances, indicating that their predictive accuracy and robustness are relatively poor. In contrast, the tree-based models—including ET, RF, BAG, ADA, and DT algorithms—demonstrated superior prediction accuracy, with the ET algorithm perform the best for MWCO and thickness, while the RF algorithm excelled for zeta potential, as schematically illustrated on the right side of Fig. 2a. Their superior performance in this specific task is consistent with the known capability of ET and RF algorithms in handling small datasets and maintaining robustness against overfitting (Gómez and Rojas, 2016; Sha et al., 2021). However, the overfitting behavior of such models was task-dependent and might differ from those observed in deep learning models.

To this end, these membrane properties were predicted using the corresponding algorithms that incorporated fabrication parameters as input features. As shown in Fig. 2b-d, a strong correlation was observed between the predicted and experimentally determined properties during the extrapolated assessment in test set, demonstrating the superior accuracy of these models.

Additionally, Shapley additive explanations (SHAP) method was used to analyze the influence of fabrication parameters on intrinsic membrane properties. The Shapley value of a feature quantifies its contribution, whether positive or negative, with a higher absolute Shapley value indicating a greater impact on membrane performance. Fig. 2e-g illustrate the top-10 most important preparation parameters (ranked from top to bottom) and their effects on membrane properties. Notably, heat curing time (HC Time), monomer A concentration (MAC), and polymerization time (PT) were identified as three critical preparation parameters for predicting membrane MWCO (Fig. 2e). Among these, heat curing time emerged as the most important fabrication parameter, exhibiting a negative correlation with MWCO across the board. This was followed by monomer A concentration and polymerization time. To elucidate the underlying mechanism of this relationship, it is essential to consider the role of heat curing in the membrane fabrication process. Heat curing plays a pivotal role in finalizing the polymerization process and removing residual organic solvents from the polyamide layer (Liu et al., 2012). During the curing process, monomers in aqueous phase migrate to the polymerization interface, thereby increasing the extent of crosslinking within the thin polyamide film (Yu et al., 2011). Prolonged heat curing durations generally enhance the crosslinking density of the polyamide active layer in the synthesized membrane (Ali et al., 2020). Furthermore, higher monomer A concentration and extended polymerization time significantly enhance the degree of crosslinking. Consequently, an increased degree of crosslinking leads to smaller pore sizes, thereby reducing the membrane's MWCO.

Shapley analysis revealed that the zeta potential exhibits a negative correlation with heat curing time and a positive correlation with polymerization time across the board (Fig. 2f). This relationship is attributable to the hydrolysis of unreacted acyl chlorides of trimesoyl chloride, which predominantly influences the zeta potential. As heat curing time and polymerization time are extended, the degree of crosslinking intensifies and the polymerization reaction progresses more completely, resulting in an increasingly positive zeta potential.

Regarding the thickness of polyamide layer, Shapley analysis identified MA1T-7, SMT-2, and MA1T-10 as the primary determinants (Fig. 2g). As described in the data collection and data set construction section, the category features were first converted into numerical features. Here, MA1T-7 stands for the encoder of the category feature (i.e., the concentration of monomer A). Specifically, MA1T-7, which corresponds to the polyethyleneimine (PEI) monomer, was found to negatively impact membrane thickness. In contrast, monomer A1 type-10 (piperazine, PIP) showed a positive correlation with polyamide membrane thickness. This discrepancy can be explained by the molecular characteristics of the monomers: PEI monomers possess larger and more complex molecular structures compared to PIP. The smaller size of PIP molecules facilitates their fast diffusion from the aqueous phase into the hexane phase and enables them to permeate the nascent polyamide film more easily, thereby reacting more readily with trimesoyl chloride and contributing to the formation of a thicker polyamide layer. Conversely, the bulkier PEI molecules are hindered by the developing polyamide film, resulting in a thinner polyamide layer (Shi et al., 2025). Additionally, substrate membrane type-2 (polyethersulfone) significantly influenced the thickness of the polyamide layer. This effect is likely due to variations in the substrate's pore structure, hydrophilicity, and surface roughness, which can affect the deposition and growth of the polyamide film during fabrication.

3.3. Development of a robust prediction model for membrane permeability and selectivity

To develop a robust prediction model for separation performance, the optimal inputs for predicting polyamide membrane Li/Mg separation efficiency were systematically investigated, encompassing membrane intrinsic properties, experimental conditions, fabrication parameters, and single-salt rejection. Previous studies commonly utilize the intrinsic properties of membranes and experimental conditions as inputs to predict membrane separation performance (Lu et al., 2023; Yong et al., 2024b). The results (Fig. 2) presented above suggest that fabrication parameters can effectively predict membrane intrinsic properties, leading to the reasonable assumption that these parameters may also serve as inputs for predicting membrane Li/Mg separation performance, as demonstrated in prior research that directly correlated monomer synthesis with membrane performance (Barnett et al., 2020; Yang et al., 2022). Moreover, a significant monotonic correlation between single-salt rejection and Li/Mg selective separation was identified (Fig. 1i), highlighting the potential to enhance the accuracy of Li/Mg selective separation prediction models when incorporated as an input.

The influence of membrane intrinsic properties, in conjunction with various experimental conditions, on membrane permeability and selectivity was initially investigated. As illustrated in Fig. 3a, the permeation performance of the membranes exhibited strong predictive capability, evidenced by Spearman correlation coefficients of 0.81. In contrast, the correlation between predicted and experimental selectivity was comparatively weaker, with a Spearman correlation coefficient of 0.65 (Fig. 3b). Fabrication parameters, which effectively predict membrane intrinsic properties, should incorporate more information than membrane intrinsic properties. Therefore, fabrication parameters were further considered as input variables, alongside experimental conditions. While the prediction performance improved, with Spearman correlation coefficients of 0.87 (Fig. 3c) and 0.75 (Fig. 3d) for permeability and selectivity, respectively, the R^2 values remained low. These findings suggest that membrane selectivity may be influenced by factors beyond the inherent membrane properties, the experimental conditions, and fabrication parameters initially evaluated.

Regarding the strong monotonic correlation between single-salt rejection and Li/Mg selective separation, the influence of single-salt rejection on membrane selectivity was examined. Consequently, selectivity in this scenario is typically assessed by evaluating the differential rejection capability of the target ions. As depicted in Fig. 4, there was strong correlation between Li/Mg selectivity and $MgCl_2$ rejection, with a Spearman coefficient of 0.74 (Fig. S4a). In contrast, no significant correlation was observed between Li/Mg selectivity and LiCl rejection, as indicated by a Spearman coefficient of 0.05 (Fig. S4b). This result suggests that $MgCl_2$ rejection capability plays an essential role in Li/Mg selectivity prediction. To further assess the potential between Li/Mg separation efficiency and single-salt rejection, two variables were selected as input features for ML: $MgCl_2$ rejection, and combined $MgCl_2$

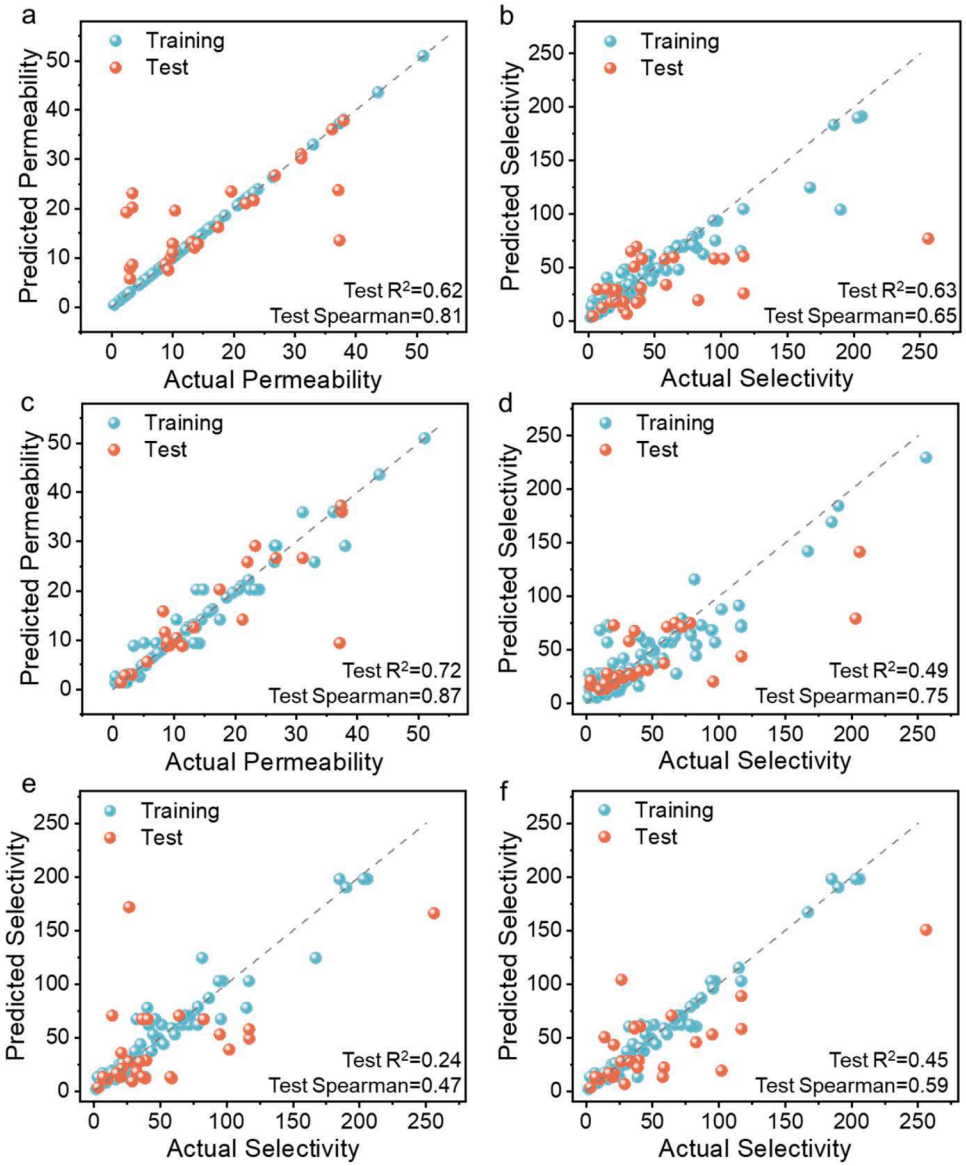


Fig. 3. (a-b) Prediction performance of the ET algorithm for (a) membrane permeability and (b) membrane selectivity based on membrane properties and experimental conditions. (c-d) Prediction performance of the (c) ET algorithm for membrane permeability and (d) BAG algorithm for membrane selectivity based on fabrication parameters and experimental conditions. (e-f) Prediction performance of the ET model for Li/Mg selectivity based on (e) $MgCl_2$ rejection and (f) combined $MgCl_2$ and LiCl rejection. All five random seed plots for the prediction performance are provided in Figs. S5–10.

and LiCl rejection. As shown in Fig. 3e, there was no significant correlation between actual selectivity and predicted selectivity when $MgCl_2$ rejection was used as input variables. This result could be attributable to the complex interactions present in mixed salt solutions, which affect Li/Mg selectivity. In such environments, the coexistence of multiple ion species can lead to competitive adsorption and charge screening effects that are not accounted for in single-salt evaluations. Specifically, the selective rejection of a particular ion by the membrane may result in the permeation of its counterions to maintain electroneutrality, thereby facilitating the transport of less impeded co-ions such as Li^+ or Mg^{2+} (Nicolini et al., 2016; Yaroshchuk, 2008). Additionally, the elevated ionic strength in mixed salt solutions enhances the screening of the membrane's charge, diminishing the electrostatic exclusion of co-ions and thereby promoting their transport (Wang et al., 2022). These interactions lead to different ion rejection behaviors in mixed salt compared to single-salt solutions, rendering single-salt rejection metrics inadequate for accurately predicting the Li/Mg selectivity. Conversely, incorporating both $MgCl_2$ and LiCl rejection data as input variables resulted in predicted selectivity for Li/Mg that better aligned experimental observations (Fig. 3f). These findings suggest the necessity of developing an ML model with salt rejection included as input feature.

To further improve the prediction accuracy, it is essential to appropriately combine the aforementioned variables. Accordingly, four sets of variables—fabrication parameters (denoted as F), experimental conditions (denoted as E), membrane properties (denoted as P), and salt rejection (denoted as S)—were integrated into the input data. Both experimental conditions and membrane properties were consistently included, while fabrication conditions and salt rejection were alternately included either individually or collectively. Subsequently, the permeability and Li/Mg selectivity of the membranes were evaluated. As displayed in Fig. 4a, regardless of the combination of input variables, the membrane's permeation performance consistently demonstrated strong predictive capability, with Spearman correlation coefficients of approximately 0.75, suggesting satisfactory prediction performance of the built ML models with membrane properties and experimental conditions as input features. However, the prediction of selectivity was

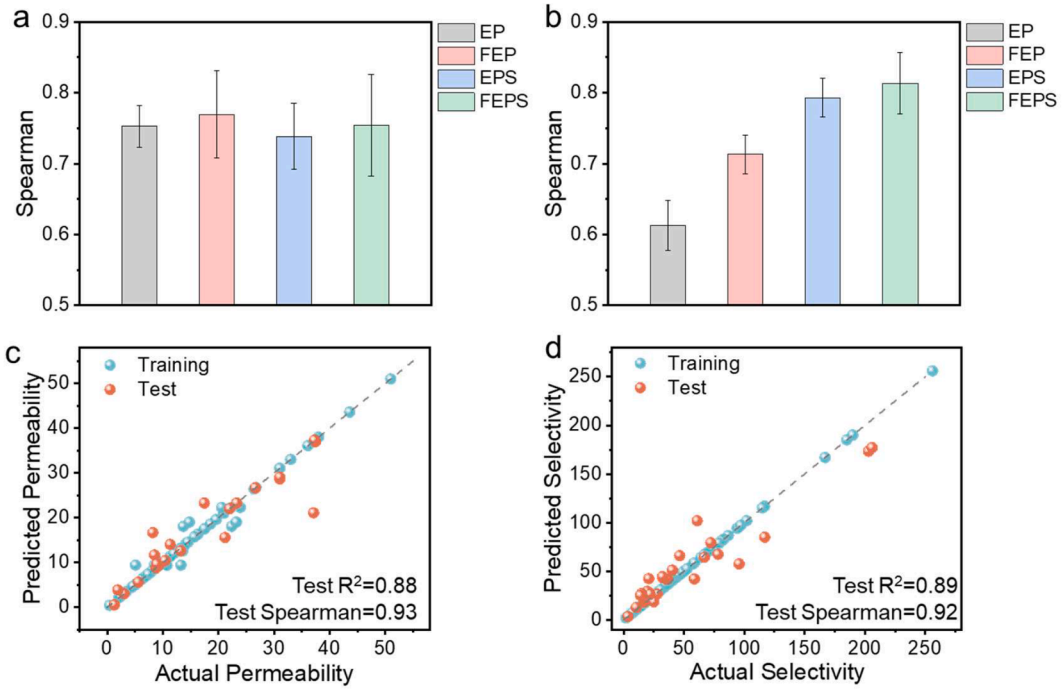


Fig. 4. Evaluation of prediction accuracy of membrane (a) permeability and (b) Li/Mg selectivity using different sets of input features. E, F, P, and S denote experimental conditions, fabrication parameters, membrane properties, and salt rejection, respectively. (c-d) Prediction performance of the ET algorithm for membrane (c) permeability based on experimental conditions, fabrication parameters, and membrane properties, and (d) Li/Mg selectivity based on experimental conditions, fabrication parameters, membrane properties, and salt rejection. The evaluation of the predictive performance of eleven ML algorithms for membrane permeability and selectivity was provided in Figs. S11–12 with Spearman correlation coefficients, as well as the MAE and RMSE values for each model in Table S3. All five random seed plots for the prediction performance are provided in Figs. S13–14.

significantly influenced by the combination of input variables (Fig. 4b). Specifically, when the input variables included only experimental conditions and membrane properties, the Spearman correlation coefficient for selectivity prediction was 0.6. When fabrication conditions were additionally incorporated into the input variables, the Spearman correlation coefficient for selectivity prediction improved to 0.7. Notably, the inclusion of salt rejection capability as an input variable further enhanced the model's predictive performance, achieving a Spearman correlation coefficient of 0.8. These findings highlight that accurate prediction of membrane selectivity necessitates a multifaceted approach that accounts for both intrinsic membrane properties and extrinsic factors such as fabrication parameters and detailed salt rejection behaviors. Future research should continue to explore and integrate additional variables to further refine predictive models, thereby advancing the design and optimization of high-performance polyamide membranes for targeted ion selective separations.

Based on the optimized combination of input features, a model capable of accurately predicting membrane mixed-salt separation performance was developed. As shown in Fig. 4c, the model achieved an R^2 of 0.88 and a Spearman coefficient of 0.93 for permeation prediction, while an R^2 of 0.89 and a Spearman coefficient of 0.91 were obtained for selectivity prediction (Fig. 4d). This result underscores the accuracy of the model with the optimized combination of input features.

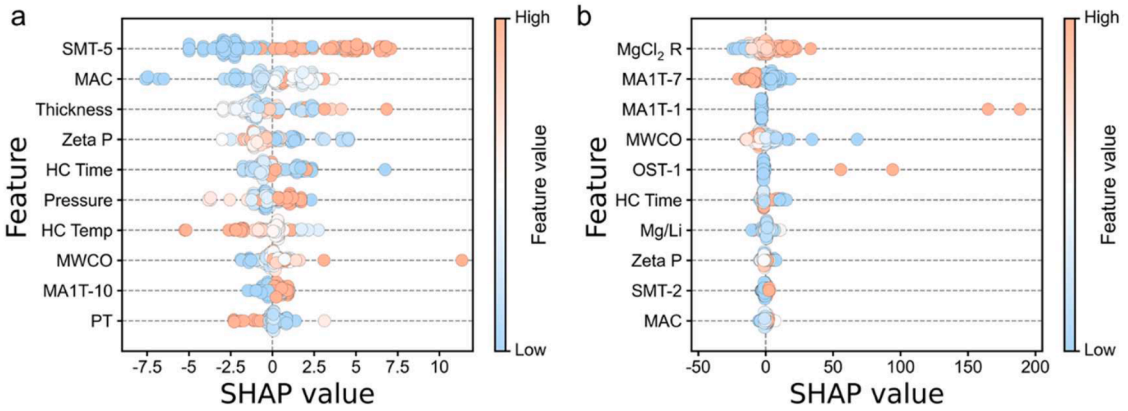


Fig. 5. (a-b) SHAP summary plot of the ET model for membrane (a) permeability based on experimental conditions, fabrication parameters, and membrane properties, and (b) Li/Mg selectivity based on experimental conditions, fabrication parameters, membrane properties, and salt rejection. HC Time, MAC, PT, HC Temp, Zeta P, $MgCl_2$ R, and Mg/Li represent heat curing time, monomer A concentration, polymerization time, zeta potential, $MgCl_2$ rejection, and the ratio of $MgCl_2$ to LiCl in the feed solution for selectivity measurement, respectively. MA1T-number, OST-number, SMT-number denote the encoders for the categorical feature of the type of monomer A1, the type of organic solvent, and the type of substrate membrane, respectively.

3.4. Feature analysis using the SHAP method for membrane permeability and selectivity

The importance and impact of each input feature to the membrane permeation performance were evaluated using the SHAP method (Fig. 5a). In general, substrate membrane type (SMT-5), MAC, thickness, and zeta potential were found to be the four most influential fabrication parameters for predicting membrane permeability. The SHAP plot demonstrated that SMT-5 positively contributed to water permeability, which means that the polysulfone might be a desirable substrate for enhancing water permeability. To optimize the robustness of the ML models and gain more insights into the effects of the polysulfone substrate, future work may focus on compiling data sets with structural or morphology parameters of the substrate, such as pore size, porosity, and thickness of substrate, surface roughness and topography of substrate, as well as its intrinsic properties (e.g., degree of sulfonation and crosslinking density).

Monomer A concentration, representing amine content, is conventionally expected to influence crosslinking density and membrane compactness—with a higher concentration typically leading to a denser structure and lower permeability. However, SHAP analysis revealed no strong monotonic correlation with membrane permeability. Interestingly, low amine concentrations were associated with reduced water permeance, suggesting that insufficient monomer availability could impair selective layer formation. Selective layer thickness showed similarly weak correlation with water flux. Notably, some membranes with greater thickness exhibited higher permeability, possibly due to the formation of favorable surface structures (e.g., hierarchical or Turing-like morphologies) (Habib and Weinman, 2021; Tan et al., 2018) that are not captured by average thickness alone. Zeta potential also positively correlated with permeability, with more negative values favoring water transport, in consistent with a previous study (Xu et al., 2025). This might be attributed to the enhanced hydration layer formation under negatively charged conditions.

Among thermal processing parameters, curing temperature showed a positive effect on permeability, while curing time did not show a clear correlation with membrane performance. This result suggests that curing temperature, rather than duration, played a more decisive role in tuning membrane morphology and facilitating effective polymer crosslinking. Polymerization time (PT) generally reduced permeability, potentially due to additional densification. Certain monomer types (e.g., MA1T-10) were associated with enhanced water flux, likely through improved hydrophilicity or crosslinking chemistry. Notably, these features ranked only 7th to 10th in SHAP importance. Such reduced contribution was likely due to collinearity, as these variables collectively influenced key structural features such as zeta potential and layer thickness (Fig. 2f-g). Taken together, these findings suggest that while membrane property parameters such as thickness and zeta potential were traditionally regarded as key indicators of transport performance, they might fail to fully capture the complex structural and interfacial features that govern water permeance. In contrast, fabrication-related parameters inherently encoded a broader set of process–structure relationships, including crosslinking kinetics, morphological heterogeneity, and surface evolution, which might indirectly reflect in the final membrane performance.

We further evaluated the contribution of each input feature to the membrane permeation performance. Among the input features, $MgCl_2$ rejection ($MgCl_2$ R), identified as the most influential factor, was positively correlated with Li/Mg selectivity (Fig. 5b). This makes sense, as both $MgCl_2$ and LiCl share the same anion in Mg/Li separation, with steric hindrance being the dominant factor (Wang and Lin, 2021). Steric hindrance is primarily determined by the pore structure of the membrane, including pore size and pore size distribution, which dictates the membrane's ability to discriminate between ions based on their physical dimensions. Therefore, $MgCl_2$ rejection could serve as an indirect indicator of a more uniform sub-nanometer pore structure. Previous studies have also shown that regulating the pore size distribution of polyamide membranes to achieve more uniform pores enhances $MgCl_2$ rejection capabilities (Liang et al., 2020; Peng et al., 2024; Shen et al., 2022). However, MWCO, a conventional metric of membrane pore size, ranked fourth only in importance, and its correlation with selectivity was weaker than that of $MgCl_2$ R based on feature importance. This result suggests that ion selectivity could not be fully explained by steric effects alone. Donnan exclusion has also been recognized as a key mechanism in ion separation, yet its commonly used proxy—zeta potential—contributed minimally in SHAP analysis. This finding, consistent with the previous studies (Jia et al., 2025; Zhao et al., 2025), indicates that zeta potential, typically measured in dilute background electrolytes such as 0.1 M KCl, fails to accurately reflect pore-localized charge or the interfacial electrostatics under high ionic strength conditions relevant to Li^+/Mg^{2+} separation. Notably, both MWCO and zeta potential were determined by membrane fabrication conditions (Fig. 2), which alone failed to accurately represent the intrinsic physicochemical properties governing ion separation. These results indicate that $MgCl_2$ rejection might serve as an integrative and informative descriptor, capturing both pore structure and surface electrostatic effects. It stands out as a performance-relevant metric that robustly bridges fabrication inputs with separation behavior, enabling more reliable prediction and interpretation of membrane selectivity.

The second most important feature, MA1T-7, which represents the PEI monomer based on the category features, negatively impacted Li/Mg selectivity. Most polyamide membranes fabricated with PEI usually exhibited a positively charged membrane surface. In binary salt solutions of $MgCl_2$ and LiCl, Mg^{2+} and Li^+ ions exhibit markedly different mass transfer behaviors. Polyamide nanofiltration membranes with highly uniform sub-nanometer pores effectively reject Mg^{2+} through size-based separation. However, the transport of Li^+ is minimally hindered by pore size due to dehydration effects, making it more dependent on membrane charge and chemical interactions. Negatively charged polyamide nanofiltration membranes enhance Li^+ permeance by promoting the transport of Cl^- counterions, whereas highly positively charged membranes increase Li^+ rejection through electrostatic attraction and Donnan exclusion (Epsztein et al., 2020; Peng et al., 2024). Consequently, negatively charged polyamide nanofiltration membranes with uniform pore distributions are more effective for extracting Li^+ from brine, as positively charged membranes undesirably elevate Li^+ rejection. Despite the theoretical role of surface charge, zeta potential was not identified as a critical feature in SHAP analysis. This reinforces the notion that zeta potential lacks spatial resolution (Zhao et al., 2025) and may fail to capture interfacial electrostatics relevant to ion partitioning and transport. Taken together, these findings underscore the need to consider both pore structure and electrostatic properties—and their combined manifestation through performance metrics like $MgCl_2$ rejection—rather than relying solely on fabrication-derived indicators, to rationalize and predict selective ion transport in nanofiltration membranes.

In support of this result, several fabrication-related variables—such as organic solvent type, substrate type, monomer concentration, and heat curing time—while ranked within the top 10 features, exhibited relatively weak influence on Li/Mg selectivity. For example, the use of *Isopar E* as a solvent appears to enhance selectivity in a few cases, but the limited representation of this condition in the dataset precludes statistically robust conclusions. Additionally, fabrication factors that favor high water permeability, such as polysulfone substrates or heat curing temperature, do not necessarily enhance selectivity. This divergence further underscores the complexity of membrane design, where permeability and selectivity are influenced by distinct factors.

Future research should focus on expanding the datasets to include detailed structural and morphological parameters of the substrate membranes, such as pore size, porosity, thickness, surface roughness, and intrinsic properties like the degree of sulfonation and crosslinking density. Incorporating these parameters could enhance the robustness of

the ML models and provide deeper insights into the effects of substrate characteristics on membrane performance. Additionally, exploring the effects of different monomer types and concentrations on membrane properties could further refine the predictive models.

Investigating the molecular-level interactions in mixed salt solutions may also provide a better understanding of the factors affecting selectivity, enabling the development of membranes with superior performance for specific separation tasks. Emphasizing the integration of comprehensive salt rejection data and advanced modeling techniques will be crucial in achieving more accurate predictions of membrane selectivity. Such advancements could significantly impact industrial processes, particularly in areas like lithium extraction from brine, by informing the design of membranes that optimize both permeability and selectivity.

4. Conclusions

This work demonstrates the effectiveness of ML models in predicting critical intrinsic properties of polyamide membranes—namely MWCO, thickness, and zeta potential—based on fabrication parameters. By integrating these membrane properties with experimental conditions, the permeability of polyamide membranes can be accurately predicted, highlighting the robust predictive capability of the developed models for permeation performance. However, predicting membrane selectivity solely based on membrane properties and experimental conditions proved inadequate. Incorporating salt rejection capability alongside these factors significantly enhanced selectivity predictions (R^2=0.89, Spearman coefficient=0.91). SHAP analysis revealed that salt rejection capability was the most influential feature in selectivity prediction. However, models incorporating $MgCl_2$ rejection as a feature should be interpreted with caution, as it represents a performance-derived metric rather than an intrinsic membrane property. These findings emphasize the necessity of a multifaceted approach in modeling membrane performance, where both intrinsic membrane properties and extrinsic factors such as fabrication parameters and salt rejection behaviors are meticulously considered. The superior performance of the ML models in predicting selectivity, when comprehensive input features are utilized, underscores the complexity of selective ion transport mechanisms and the importance of capturing detailed membrane characteristics.

Data and code availability

Data are provided in the supporting Excel file. Code will be made available on request.

CRediT authorship contribution statement

Jing-Ou Sun: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Tian-Wei Hua:** Writing – original draft, Validation, Software, Formal analysis, Data curation, Conceptualization. **Yan-Fang Guan:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Han-Qing Yu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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